Determinants of the Number of Kids of Parents in the US

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Introduction

For the project, the team originally wanted to answer a specific question: what is the maximum number of kids a family should have based on certain family-related variables? However, due to limitations with the available data, the team pivoted to focus on how certain family variables affect the number of children a family decides to have. While this study is ultimately focused on understanding how certain variables affect the number of kids a couple has, this study could be used as the foundation to later identify the optimal number of children that people in the US should have based on a variety of potential inputs. These questions are relevant to overall society and putting a concrete answer to the questions would help people avoid financial distress, while encouraging better environments for children. Understanding these effects will prevent families from having too many children too early, leading to poor quality of life, poverty and general unhappiness. Government officials could exploit this knowledge to aid pre-existing families with too many children.

Ultimately, another project could focus on creating a model to optimize the number of kids a family could have. For example, an individual or couple could input their income, age, savings, residential address, job type and other factors into a model, which would then let the potential parents know their personal capacity for having children. As mentioned, having this model accessible will not only limit the amount of kids that a couple might have, but potentially encourage families to grow, by indicating that the parents are stable enough to have more children.

In order to address this question, the team needed to find data broken down by geographical location. At the state or county level, the team wanted to find average cost of living, average cost of having a kid (from ages 0-21), job security, tax information, accessibility

to government assistance programs and more. The team looked for cross-sectional data to run regressions on the aforementioned information to understand how factors like this affect the number of children in a family. Ultimately, another team or research project could expand on this information to solve for the optimal number of children a family could have. Then, they could make a simple calculator from this project that will be available to the public for free use.

Literature Review

An overall assessment of the literature as a whole has revealed that extensive research has been done on the topic of family size and its relation with family welfare, academic achievement, income and consumption, etc. However, the scope of the discussion focuses extensively on family size in developing regions of the world as opposed to the United States, and in-family differences such as birth order affects children.

The existing body of research does not include an in-depth analysis of how different variables influence family size or a predictive model to consult the optimal number of children to have based on different factors. So, the team hoped that their contribution to the discussion could ignite scholars to engage in the research with the aim of informing the public and government entities of family planning related procedures and policies.

Resource dilution model has been a prevalent topic discussed in the literature about its links to children's welfare. Black, Devereux, and Salvanes (2005) aimed to examine the tradeoff between child quantity and quality within a family. They found that the effect of family size had a negative impact on educational attainment but when confounded with the birth order, the correlation is nullified. They also studied adult earnings, employment, and teenage childbearing and discovered a strong correlation with birth order in females, implicating the need to include in-family differences for children. This paper provides a foundational understanding as to how

family size and the birth order affect the child's education. Since common sense dictates a strong correlation between education and success, this would be applicable to the topic.

Sandfail and Rafail's (2014) study measured the effect of family size on children's educational attainment. The research also pointed out that children's time spent with parents in household or extracurricular activities is a key reason behind the confluence and the resource dilution model. Nevertheless, the results of the cross-sectional analysis present relatively weak support for both confluence and resource dilution models. Again, since the correlation of educational performance and success is presumed to be high, this paper would suggest that family size and the family's ability to spend time with each other dictate child success. These could have become variables in the team's regression model.

Miller and Amalia (2011) investigate the effects of motherhood timing on career paths. Delaying motherhood was found to have substantial increases on earnings(9%), wages(3%), and work hours(6%). Additionally, the impact of delaying motherhood was found to be more significant for college-educated women in professional positions. This paper served to help the team better understand the timing of having children for a household. The paper discussed several terms that could prove useful for further research such as fertility timing, and the 'mommy track', "the effect of a 1-year career interruption associated with motherhood at ages 23 and 30". Additionally, besides the literal contents of the paper, this source provided an example of effective incorporation of formulas and visualizations.

Chetty, Hendren, Jones, and Porter (2020) studied economic sources of racial disparities. The paper cites three major findings: "black Americans and American Indians have much lower rates of upward mobility and higher rates of downward mobility than whites"(1), "differences in parental marital status, education, and wealth explain little of the black-white income gap

conditional on parent income"(1), "the black-white gap persists even among boys who grow up in the same neighborhood. Controlling for parental income, black boys have lower incomes in adulthood than white boys in 99% of Census tracts"(1). This source helped the team understand a potential confounding variable in the research topic: racial disparities. Failing to control for race in the regression could lead to omitted variable bias. Thus, it was important to find a paper to better contextualize racial disparities impact on income.

Cáceres-Delpiano (2006) discusses the effects that family size and investment has on the quality of a child's life. Quality is measured through academic achievement, future wages, and labor force participation. In order to measure investment, Cáceres-Delpiano looks at many different factors. He then uses these variables to calculate the quality of life for children, and how the number of kids affects the quality. The data suggests that although there is a higher cost for more kids, parents are able to reallocate where they invest their time and money, mitigating the effects on their children's quality of life. Cáceres-Delpiano initially believed that larger families would lead to much less time devoted to children. However, parents seem to understand the responsibility accosted with more children, and are able to make it work for the betterment of their children. This paper greatly helped the team's research as it gave an insight into the variables they looked at, such as private school and time spent working, and correlates it with quality of life for children. The team was able to delve into what makes each of the variables important. Additionally, it was interesting to look further into parents' ability to reallocate time and resources, which is something the team had not discussed much until reading the article. Parents can typically figure out a way to make everything work, so the team would like to analyze more what adjustments need to be made as they increase their family size.

Mikko and Margolis (2014) dive deep into the relationship between well-being

(happiness) and having kids. Using large British and German longitudinal data sets, the authors observed when parents are the happiest before and after having kids and determined how many kids make the parents the happiest. The study also found that having two kids maximizes happiness, and having more children didn't make parents happier. Furthermore, the study finds that women tend to be happier than men when having kids, those married tend to be happier than those unmarried when having kids, those who are higher educated tend to be happier than those whole are less educated when having kids, and older parents tend to be happier than younger parents when having kids. Also net economic cost does not vary too much between having a second and a third kid. So, the authors argue that the optimal number of children is two. The fact that Myrskylä and Margolis argue that having two kids maximizes the happiness of parents is super beneficial in answering the question. Well-being (happiness) is an extremely important factor to consider when having kids. In fact, the authors argue that it is even a more influential factor than economic factors. It is really interesting that the author argues the economic factor isn't the primary factor on why parents are having fewer kids since it contradicts with some of the other sources, thus, this article can potentially be a great reference for building a juxtaposition to the team's thesis.

Hypothesis

The team aims to investigate how extraneous variables influence the number of kids per parent in the US. The dependent variable is the number of children (newborns to age 17) in a family unit. Among the independent variables, the team hypothesized that the effect of income in USD on the number of children is positive. In other words, one should expect an increase in the number of children as income level increases by a positive amount. In the context of the current literature surrounding the factors affecting family size, this study on the effect of income in the

United states on the number of children serves to update the findings in previous papers (Woofter, 1944 and Cutright, 1971). This study also aimed to highlight the effects of race and sexual orientation that is absent and overlooked in the present literature regarding income and family size.

While the literature mainly deals with how all the variables relate with one another, the team will look into how each of the variables plays into the number of kids a family has. Each of the articles discussed deals with separate variables and does not combine everything to see what has a bigger effect on the number of kids. For example, this study hopes to see how the effect of geographic location on the dependent variable compares to income. These insights will add to the already great research done previously, as the team can essentially find the trends in the number of kids, and where and when populations are likely to grow or decline.

Methodology

The data that will be used to conduct this study was taken from the Panel Study of Income Dynamics (PSID). The cross-sectional data is taken from 2019, the most recent year with available data. The data consists of the following columns: a distinct family interview identification number, the state in which the interviewee resides, the age and sex of the interviewee and their spouse, the number of people in their family, their yearly salary or income amount, the education level of the parents and their race. Ultimately, this data will be used to regress the number of kids in a family with the other exogenous variables. The data was collected via interviews and the unit of observation for the data is the distinct family interview identification number. This data has many strengths: it is from a trusted source, it contains over nine-thousand observations, it contains a lot of exogenous variables and it details a wide-array of information. However, the only substantial weakness of the data is that it does not consider

whether or not the family is done having children. However, this should not significantly affect the project's ability to answer the question of -- how do certain exogenous variables affect the expected number of kids in a family -- because other variables like age and income should account for this omitted variable.

Variables and Econometric Model

The regression's dependent variable will be the number of kids in the family. The independent variables will be the state in which the interviewee resides, the age and sex of the interviewee and their spouse, their yearly salary or income amount, the education level of the parents and their race. The dependent variable, or the number of kids in the family, will be used to measure the effect of changing an exogenous variable on family size. The team expects each of these variables to have a significant effect on the number of kids, because each variable demonstrates a strong link to a person's ability to care for children.

Since there is a great cost to raising children, it is assumed that income will significantly affect family size. So, the regression model will ultimately explain the relationship between income and family size, which will have great implications in the model. If income is positively related to family size, then it will be known that lower income families are disproportionately able to have more children. Furthermore, it will be evident if educated parents make more sound decisions based on their economic stability. Their income will likely have a stronger correlation with the number of kids compared to an uneducated person, as someone with a degree is more inclined to make more pragmatic decisions.

Race is another key independent variable that will hopefully explain differences in family size due to race. In the regression, there will be dummy variables for each of the races used in the dataset. This will ultimately identify which races are more likely to have larger families.

Geographical location will also influence family size, as cultural differences from state to state will likely illustrate the changes in beliefs. Since the fifty states are nominal data, entity-fixed effects must be used to measure this difference. With the entity-fixed effects by state data, the team could look at which geographic locations across America tend to have more children, and see how it correlates with income. These effects will demonstrate how cultural trends shift as one travels to different places, even if it is still the same country.

Technique - An OLS regression will be modeled to test the hypothesis based on its fit with on dependent variable, a non-binary, non-panel, cross-sectional dataset. OLS is the best model for the data because all the variables are exogenous (no perfect multicollinearity), and that errors are assumed to be homoscedastic and normally distributed over the sample size of 8650 valid observations.

Specification - The model could potentially suffer from omitted variable bias as the parameters of the dataset are limited to certain characteristics asked in the interview. Moreover, income and number of children could suffer from simultaneous causality which might indicate that instrumental variable regression could be implemented if that is the case. The team will also investigate the potential for interaction amongst variables such as sexuality of households and states that the family resides in due to state regulations.

Validity of claim - The use of heteroskedastic-consistent robust standard errors will be used in the model. There is no need to control for autocorrelation and utilize clustered standard error as correlation within each entity is absent in cross-sectional data. Adjusted R^2 will be used to evaluate imperfect multicollinearity and the effectiveness of different models in explaining the variations in the number of children.

Types of statistical tests - Individual t-tests will be used to validate the statistical significance of the effect of independent variables on the number of children. F-test will be used to determine whether the interaction term between two independent variables is of statistical significance.

Econometric Model

This paper uses a linear regression model to depict statistically significant cross-sectional variables and its ability to explain the variation in the number of children. The econometric equation derived from the fixed effects by state OLS method is shown below:

$$\text{NUM_CHILD} = \beta_0 + \beta_1\text{AGE} + \beta_2\text{SEX} + \beta_3\text{S_AGE} + \beta_4\text{S_SEX} + \beta_5\ln(\text{Salary}) + \beta_6\text{FATHER.EDUC} + \beta_7\text{S_RELIGION} + \beta_8\text{RACE} + \beta_9\text{RELIGION} + \beta_{10}\text{SAMESEX} + \beta_{11}\text{Age40orOlder} + \beta_{12}\text{Age60orOlder} + \beta_{13}\text{S_40orOlder} + \beta_{14}\text{S_50orOlder} + \beta_{15}\text{S_60orOlder} + \beta_{16}(\text{AGE} * \text{Age40orOlder}) + \beta_{17}(\text{AGE} * \text{Age60orOlder}) + \beta_{18}(\text{S_AGE} * \text{S_Age40orOlder}) + \beta_{19}(\text{S_AGE} * \text{S_Age50orOlder}) + \beta_{20}(\text{S_AGE} * \text{S_Age60orOlder}) + \textit{StateFixedEffects}$$

Empirical Analysis and Findings

The model uses a fixed effects model to encapsulate state specific effects and thus eliminate potential omitted variable bias. The model went through several iterations. Initially the team tried to perform a fixed effects regression by states with salary as the only explanatory variable, as seen in Model 1 [Appendix D]. While this model presented a relatively strong adjusted R^2 of 0.3068, there was certainly omitted variable bias as, logically speaking, salary is not the only determinant of the number of children. The next iteration included salary as the only explanatory variable, but in an attempt to provide a more logical interpretation of the coefficient, the team performed a logarithmic transformation and ran a fixed effects regression by states with ln(Salary) as seen in Model 2 [Appendix E]. As expected, the model was nearly identical to

Model 1, with an adjusted R^2 of 0.307. However, the interpretation of the coefficient of the salary variable from Model 2 was easier to comprehend as the expected increase in number of children for a 1% increase in salary -- as opposed to a $1 increase in salary. Then, the team included the effects of race, age, religious preference, for both the reference person and spouse, in addition to the education of the father and mother. They found Model 3 [Appendix F] to be an improvement upon their salary based models, with an adjusted R^2 of 0.4556. Upon analyzing the data further and considering the relationship between age demographics and number of children, they included binary age stratifying variables for the reference person and spouse, an indicator variable for being 40 or older, 50 or older, and 60 or older, in Model 4 [Appendix G]. Ultimately, the team found the inclusion of these variables to once again improve upon the predictive power of their model, increasing the adjusted R^2 to 0.5037. Next, they included an interaction term between the age stratifying variable and the age of the person in Model 5 [Appendix H], and once again found adjusted R^2 to increase to 0.5315. Finally, the team dropped all the coefficients that were not statistically significant (mother education, race of the spouse, reference person 50 or older, and the interaction term between reference person 50 or older and reference person age). The adjusted R^2 of the final model [Appendix I] was 0.5315, meaning that 53.15% of the variation in the number of children was accounted for by the model. Finally, the team found the final model to be statistically significant with a p-value of 2.2e-16.

Conclusion

The dynamics of family planning vary geographically in the United States. The socioeconomic status of households vary by state, geographic factors also dictate attitudes towards same-sex marriage, interracial marriage, educational attainment and religious beliefs.

This paper aimed to explore the effectiveness of income and family size and pursued the difference amongst states to present a guideline for family planning programmes at a federal level and at a state level. Policy makers could benefit from empirical findings to have programs in place to help households avoid financial distress from having children too early, as well as exploit this knowledge to aid pre-existing families with too many children. Moreover, the paper broaches the discussion on the emerging inclusive, diverse and equitable environment and its impact on family planning as it traverses the impact of social factors on the optimal number of children to have.

The empirical findings of this paper is derived from the Panel Study of Income Dynamics (PSID) database 2019 annual interview. After cleaning the dataset, 8580 observations were included, each observation containing 17 characteristics ranging from number of children to religion of the spouse.

The fixed effects estimator model showed that a percentage change in salary had a statistically significant impact on the number of children in the United States in 2019 when holding variables such as age, sex, race, religion and state differences constant. Nevertheless, based on the availability of the dataset and the selected interview questions, the model potentially suffers from external validity and omitted variable bias. One such variable that was not asked during the interview was the measurement of workload such as hours per week at work or overtime hours. As previous literature has suggested, controlling for workload captures the effect of motherhood timing on career path (Miller and Amalia, 2011), Resource dilution (Black, Devereux, and Salvanes, 2005) and happiness of children (Mikko and Margolis, 2014).

In addition, an extension of the paper would be to use a MLE estimator to predict the probability or sensibility for a household with the given characteristics to have their first child.

With the binary dependent variable being one if the existing factors suggest that the family should have a kid and zero otherwise.

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Appendix A: Variable Definitions

ER72004 "CURRENT STATE"

Current State (FIPS Code)

Please refer to FIPS state codes here http://psidonline.isr.umich.edu/data/Documentation/FIPSStateCodes.pdf

Count	%	Value/Range	Code Value/Range Text
9,519	99.48	1 - 56	Actual state (FIPS code)
-	-	99	DK; NA
50	.52	0	Inap.: U.S. territory or foreign country

ER72017 "AGE OF REFERENCE PERSON"

Age of 2019 Reference Person

This variable represents the actual age of the 2019 Reference Person of the FU. The minimum value is usually 18, although in rare cases a person under 18 might become Reference Person.

Count	%	Value/Range	Code Value/Range Text
9,568	99.99	14 - 120	Actual age
1	.01	999	DK; NA

ER72018 "SEX OF REFERENCE PERSON"

Sex of 2019 Reference Person

Count	%	Value/Range	Code Value/Range Text

6,469	67.60	1	Male
3,100	32.40	2	Female

ER72019 "AGE OF SPOUSE"

Age of 2019 Spouse/Partner

This variable represents the actual age of the current Spouse or Partner (cohabiting friend). Spouses/Partners aged 14-16 are rare.

Count	%	Value/Range	Code Value/Range Text
4,842	50.60	14 - 120	Actual age
5	.05	999	DK; NA
4,722	49.35	0	Inap.: Reference Person is female or single male; no Spouse/Partner in FU (ER73422=5)

ER72020 "SEX OF SPOUSE"

Sex of Spouse

Count	%	Value/Range	Code Value/Range Text
18	.19	1	Male
4,829	50.47	2	Female
4,722	49.35	0	Inap.: Reference Person is single male or female; no Spouse/Partner in FU (ER73422=5)

ER72021 "# CHILDREN IN FU"

Number of Persons Now in the FU Under 18 Years of Age

This variable represents the actual number of persons currently in the FU who are neither Reference Person nor Spouse/Partner, from newborns through those 17 years of age, whether or not they are actually children of the Reference Person or Spouse/Partner.

Count	%	Value/Range	Code Value/Range Text
5,808	60.70	0	None
3,761	39.30	1 - 18	Actual number

ER72211 "BC30 SALARY AMOUNT"

BC30. How much is your salary?--AMOUNT FOR CURRENT MAIN JOB

The values for this variable represent dollars and cents.

Count	%	Value/Range	Code Value/Range Text
2,133	22.29	.01 - 9,999,996.99	Actual amount
-	-	9,999,997.00	$9,999,997 or more

27	.28	9,999,998.00	DK
77	.80	9,999,999.00	NA; refused
7,332	76.62	.00	Inap.: not currently employed (ER72164, ER72165, and ER72166 ne 1 or 2 and ER72168=5); DK, NA, or RF whether working for money now (ER72168=8 or 9); is not salaried (ER72210=3, 4, 6, or 7); DK, NA, or RF how paid on main job (ER72210=8 or 9)

ER76715 "K5 EDUCATION OF FATHER IN US-SP"

K5. How much education did (your/her/his) father complete (in the United States)?

Count	%	Value/Range	Code Value/Range Text
85	.89	1	Completed 0-5 grades
286	2.99	2	Completed 6-8 grades; "grade school"; DK but mentions could read and write
299	3.12	3	Completed 9-11 grades (some high school); junior high
1,477	15.44	4	Completed 12 grades (completed high school); "high school"
210	2.19	5	Completed 12 grades plus nonacademic training; R.N. (no further elaboration)
433	4.53	6	Completed 13-14 years; Some college, no degree; Associate's degree
563	5.88	7	Completed 15-16 years; College BA and no advanced degree mentioned; normal school; R.N. with 3 years college; "college"
319	3.33	8	Completed 17 or more years; College, advanced or professional degree, some graduate work; close to receiving degree
362	3.78	99	DK; NA; refused
5,535	57.84	0	Inap.: no Spouse/Partner in FU (ER73422=5); educated outside the U.S. only or had no education (ER76714=2 or 5); NA or RF where father received his education (ER76714=9)

ER76725 "K15 EDUCATION OF MOTHER IN US-SP"

K15. How much education did (your/her/his) mother complete (in the United States)?

Count	%	Value/Range	Code Value/Range Text
42	.44	1	Completed 0-5 grades
216	2.26	2	Completed 6-8 grades; "grade school"; DK but mentions could read and write
327	3.42	3	Completed 9-11 grades (some high school); junior high
1,564	16.34	4	Completed 12 grades (completed high school); "high school"
210	2.19	5	Completed 12 grades plus nonacademic training; R.N. (no further elaboration)
631	6.59	6	Completed 13-14 years; Some college, no degree; Associate's degree

591	6.18	7	Completed 15-16 years; College BA and no advanced degree mentioned; normal school; R.N. with 3 years college; "college"
231	2.41	8	Completed 17 or more years; College, advanced or professional degree, some graduate work; close to receiving degree
206	2.15	99	DK; NA; refused
5,551	58.01	0	Inap.: no Spouse/Partner in FU (ER73422=5); educated outside the U.S. only or had no education (ER76724=2 or 5); NA or RF where mother received her education (ER76724=9)

ER76752 "K40 RACE OF SPOUSE-MENTION 1"

K40. What is (your/her/his) race? (Are you/Is [he/she]) white, black, American Indian, Alaska Native, Asian, Native Hawaiian or Pacific Islander?--FIRST MENTION

Count	%	Value/Range	Code Value/Range Text
3,280	34.28	1	White
1,097	11.46	2	Black, African-American, or Negro
38	.40	3	American Indian or Alaska Native
114	1.19	4	Asian
4	.04	5	Native Hawaiian or Pacific Islander
254	2.65	7	Other
60	.63	9	DK; NA; refused
4,722	49.35	0	Inap.: no Spouse/Partner in FU (ER73422=5)

ER76815 "K68A RELIGIOUS PREFERENCE-SP"

K68a. We'd like to ask you some questions about religious or spiritual beliefs. What is (your /SPOUSE/PARTNER's) religious preference?

IF R says Christian, PROBE for specific denomination; if R cannot provide specific denomination ENTER [14]
IF R says Protestant, PROBE for specific denomination; if R cannot provide specific denomination, ENTER [13]

Count	%	Value/Range	Code Value/Range Text
645	6.74	1	None/No Preference
74	.77	2	Atheist
60	.63	3	Agnostic
904	9.45	4	Roman Catholic
12	.13	5	Greek/Russian/Eastern Orthodox
789	8.25	6	Baptist
42	.44	7	Episcopalian
41	.43	8	Jehovah's Witness
162	1.69	9	Lutheran
197	2.06	10	Methodist; African Methodist

135	1.41	11	Pentecostal; Assembly of God
63	.66	12	Presbyterian
105	1.10	13	Protestant, Unspecified
703	7.35	14	Christian, Unspecified
411	4.30	15	Christian, Non-denominational
83	.87	16	Jewish (Judaism)
17	.18	17	Muslim (Islam)
20	.21	18	Buddhist
47	.49	19	Other non-Christian
17	.18	20	Other Protestant
57	.60	21	Latter Day Saints; Mormon
9	.09	22	Unitarian; Universalist
-	-	23	Christian Science
15	.16	24	Seventh Day Adventist
5	.05	25	Amish; Mennonite
2	.02	26	Quaker; Friends
26	.27	27	Church of God
5	.05	28	United Church of Christ; Congregational Church
10	.10	29	Reformed, Christian Reformed
24	.25	30	Disciples of Christ; United Christian; First Christian; Christian Holiness
27	.28	31	Churches of Christ
7	.07	97	Other
-	-	98	DK
133	1.39	99	NA; refused
4,722	49.35	0	Inap.: no Spouse/Partner in FU (ER73422=5)

ER76897 "L40 RACE OF REFERENCE PERSON-MENTION 1"

L40. What is (your/his/her) race? (Are you/Is [he/she]) white, black, American Indian, Alaska Native, Asian, Native Hawaiian or other Pacific Islander?--FIRST MENTION

Count	%	Value/Range	Code Value/Range Text
5,166	53.99	1	White
3,713	38.80	2	Black, African-American, or Negro
62	.65	3	American Indian or Alaska Native
141	1.47	4	Asian
13	.14	5	Native Hawaiian or Pacific Islander
391	4.09	7	Other
83	.87	9	DK; NA; refused

ER76960 "L68A RELIGIOUS PREFERENCE-RP"

L68a. We'd like to ask you some questions about religious or spiritual beliefs. What is (your /REFERENCE PERSON's) religious preference?

IF R says Christian, PROBE for specific denomination; if R cannot provide specific denomination ENTER [14]
IF R says Protestant, PROBE for specific denomination; if R cannot provide specific denomination, ENTER [13]

Count	%	Value/Range	Code Value/Range Text
1,632	17.06	1	None/No Preference
172	1.80	2	Atheist
185	1.93	3	Agnostic
1,382	14.44	4	Roman Catholic
16	.17	5	Greek/Russian/Eastern Orthodox
1,945	20.33	6	Baptist
64	.67	7	Episcopalian
92	.96	8	Jehovah's Witness
216	2.26	9	Lutheran
360	3.76	10	Methodist; African Methodist
223	2.33	11	Pentecostal; Assembly of God
76	.79	12	Presbyterian
205	2.14	13	Protestant, Unspecified
1,364	14.25	14	Christian, Unspecified
763	7.97	15	Christian, Non-denominational
126	1.32	16	Jewish (Judaism)
57	.60	17	Muslim (Islam)
36	.38	18	Buddhist
71	.74	19	Other non-Christian
23	.24	20	Other Protestant
71	.74	21	Latter Day Saints; Mormon
8	.08	22	Unitarian; Universalist
2	.02	23	Christian Science
21	.22	24	Seventh Day Adventist
5	.05	25	Amish; Mennonite
4	.04	26	Quaker; Friends
33	.34	27	Church of God
12	.13	28	United Church of Christ; Congregational Church
10	.10	29	Reformed, Christian Reformed
40	.42	30	Disciples of Christ; United Christian; First Christian; Christian Holiness

62	.65	31	Churches of Christ
21	.22	97	Other
-	-	98	DK
268	2.80	99	NA; refused
4	.04	0	Inap.: no Spouse/Partner in FU (ER73422=5)

Appendix B: Table of Descriptive Statistics for Initial[1] Data Set

```
    STATE          AGE.REFP         SEX.REFP       AGE.SPOUSE       SEX.SPOUSE      NUM.CHILD
 Min.   : 0.00   Min.   : 18.00   Min.   :1.000   Min.   :  0.00   Min.   :0.000   Min.   : 0.0000
 1st Qu.:13.00   1st Qu.: 33.00   1st Qu.:1.000   1st Qu.:  0.00   1st Qu.:0.000   1st Qu.: 0.0000
 Median :28.00   Median : 43.00   Median :1.000   Median : 22.00   Median :2.000   Median : 0.0000
 Mean   :27.99   Mean   : 46.54   Mean   :1.324   Mean   : 23.62   Mean   :1.011   Mean   : 0.7886
 3rd Qu.:41.00   3rd Qu.: 59.00   3rd Qu.:2.000   3rd Qu.: 43.00   3rd Qu.:2.000   3rd Qu.: 1.0000
 Max.   :56.00   Max.   :999.00   Max.   :2.000   Max.   :999.00   Max.   :2.000   Max.   :10.0000
    SALARY          EDU.FATHER       EDU.MOTHER      RACE.SPOUSE    RELIGION.SPOUSE    RACE.REFP
 Min.   :      0   Min.   : 0.000   Min.   : 0.000   Min.   :0.000   Min.   : 0.000   Min.   :1.000
 1st Qu.:      0   1st Qu.: 0.000   1st Qu.: 0.000   1st Qu.:0.000   1st Qu.: 0.000   1st Qu.:1.000
 Median :      0   Median : 0.000   Median : 0.000   Median :1.000   Median : 1.000   Median :1.000
 Mean   : 125253   Mean   : 5.585   Mean   : 4.068   Mean   :0.876   Mean   : 5.719   Mean   :1.765
 3rd Qu.:      0   3rd Qu.: 4.000   3rd Qu.: 4.000   3rd Qu.:1.000   3rd Qu.: 6.000   3rd Qu.:2.000
 Max.   :9999999   Max.   :99.000   Max.   :99.000   Max.   :9.000   Max.   :99.000   Max.   :9.000
 RELIGION.REFP
 Min.   : 0.00
 1st Qu.: 4.00
 Median : 6.00
 Mean   :10.89
 3rd Qu.:14.00
 Max.   :99.00
```

[1]Renamed variables for regression convenience

Appendix C: Table of Descriptive Statistics for Cleaned[2] Data Set

```
   STATE              AGE.REFP         SEX.REFP       AGE.SPOUSE      SEX.SPOUSE       NUM.CHILD
Length:8650        Min.   : 18.00   Min.   :1.000   Min.   : 0.00   Min.   :0.0000   Min.   : 0.0000
Class :character   1st Qu.: 33.00   1st Qu.:1.000   1st Qu.: 0.00   1st Qu.:0.0000   1st Qu.: 0.0000
Mode  :character   Median : 43.00   Median :1.000   Median : 0.00   Median :0.0000   Median : 0.0000
                   Mean   : 46.37   Mean   :1.343   Mean   :21.84   Mean   :0.9586   Mean   : 0.7765
                   3rd Qu.: 59.00   3rd Qu.:2.000   3rd Qu.:42.00   3rd Qu.:2.0000   3rd Qu.: 1.0000
                   Max.   :102.00   Max.   :2.000   Max.   :91.00   Max.   :2.0000   Max.   :10.0000
    SALARY         EDU.FATHER      EDU.MOTHER     RACE.SPOUSE     RELIGION.SPOUSE   RACE.REFP
Min.   :     0   Min.   :0.000   Min.   :0.000   Min.   :0.0000   Min.   : 0.00   Min.   :1.000
1st Qu.:     0   1st Qu.:0.000   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.: 0.00   1st Qu.:1.000
Median :     0   Median :0.000   Median :0.000   Median :0.0000   Median : 0.00   Median :1.000
Mean   : 17068   Mean   :1.902   Mean   :1.934   Mean   :0.7803   Mean   : 4.21   Mean   :1.694
3rd Qu.:     0   3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:1.0000   3rd Qu.: 6.00   3rd Qu.:2.000
Max.   :600000   Max.   :8.000   Max.   :8.000   Max.   :7.0000   Max.   :97.00   Max.   :7.000
RELIGION.REFP       lnSALARY         SAMESEX         x40ORolder.REFP  x50ORolder.REFP  x60ORolder.REFP
Min.   : 0.000   Min.   : 0.000   Min.   :0.000000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
1st Qu.: 4.000   1st Qu.: 0.000   1st Qu.:0.000000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
Median : 6.000   Median : 0.000   Median :0.000000   Median :1.0000   Median :0.0000   Median :0.0000
Mean   : 8.337   Mean   : 2.424   Mean   :0.005549   Mean   :0.5772   Mean   :0.3908   Mean   :0.2461
3rd Qu.:14.000   3rd Qu.: 0.000   3rd Qu.:0.000000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:0.0000
Max.   :97.000   Max.   :13.305   Max.   :1.000000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
x40ORolder.SPOUSE x50ORolder.SPOUSE x60ORolder.SPOUSE
Min.   :0.0000    Min.   :0.0000    Min.   :0.0000
1st Qu.:0.0000    1st Qu.:0.0000    1st Qu.:0.0000
Median :0.0000    Median :0.0000    Median :0.0000
Mean   :0.2739    Mean   :0.1794    Mean   :0.1031
3rd Qu.:1.0000    3rd Qu.:0.0000    3rd Qu.:0.0000
Max.   :1.0000    Max.   :1.0000    Max.   :1.0000
```

[2]Cleaned by dropping observations with DK, NA, or Refused. Added variables to be used in models

Appendix D: Regression Summary Table for Model 1 Fixed Effects by State only with Salary

```
Call:
lm(formula = NUM.CHILD ~ SALARY + factor(STATE) - 1, data = df.data)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.4868 -0.7658 -0.6684  0.3751  9.2579 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
SALARY           1.503e-06  3.040e-07   4.943 7.84e-07 ***
factor(STATE)AK  5.555e-01  3.152e-01   1.762 0.078031 .  
factor(STATE)AL  7.448e-01  1.016e-01   7.333 2.46e-13 ***
factor(STATE)AR  7.626e-01  9.667e-02   7.889 3.42e-15 ***
factor(STATE)AZ  7.658e-01  1.040e-01   7.366 1.92e-13 ***
factor(STATE)CA  7.218e-01  4.247e-02  16.995  < 2e-16 ***
factor(STATE)CO  6.846e-01  9.570e-02   7.153 9.17e-13 ***
factor(STATE)CT  5.736e-01  1.739e-01   3.298 0.000978 ***
factor(STATE)DC  7.539e-01  1.563e-01   4.825 1.43e-06 ***
factor(STATE)DE  4.827e-01  3.730e-01   1.294 0.195602    
factor(STATE)FL  6.466e-01  5.967e-02  10.837  < 2e-16 ***
factor(STATE)GA  8.421e-01  6.568e-02  12.822  < 2e-16 ***
factor(STATE)HI  1.188e-01  4.816e-01   0.247 0.805102    
factor(STATE)IA  8.922e-01  9.333e-02   9.559  < 2e-16 ***
factor(STATE)ID  1.057e+00  3.152e-01   3.353 0.000803 ***
factor(STATE)IL  6.684e-01  7.416e-02   9.013  < 2e-16 ***
factor(STATE)IN  9.396e-01  7.172e-02  13.101  < 2e-16 ***
factor(STATE)KS  1.126e+00  1.742e-01   6.465 1.07e-10 ***
factor(STATE)KY  6.767e-01  1.012e-01   6.685 2.45e-11 ***
factor(STATE)LA  8.644e-01  1.035e-01   8.354  < 2e-16 ***
factor(STATE)MA  4.565e-01  1.006e-01   4.536 5.81e-06 ***
factor(STATE)MD  7.203e-01  7.077e-02  10.178  < 2e-16 ***
factor(STATE)ME  9.637e-01  2.515e-01   3.831 0.000129 ***
factor(STATE)MI  8.168e-01  5.681e-02  14.378  < 2e-16 ***
factor(STATE)MN  8.577e-01  1.132e-01   7.580 3.82e-14 ***
factor(STATE)MO  8.498e-01  7.530e-02  11.286  < 2e-16 ***
factor(STATE)MS  7.806e-01  6.524e-02  11.965  < 2e-16 ***
factor(STATE)MT  7.991e-01  3.729e-01   2.143 0.032168 *  
factor(STATE)NC  6.968e-01  5.389e-02  12.931  < 2e-16 ***
factor(STATE)ND  1.656e-01  3.730e-01   0.444 0.657168    
factor(STATE)NE  1.239e+00  1.258e-01   9.848  < 2e-16 ***
factor(STATE)NH  7.402e-01  2.515e-01   2.943 0.003262 ** 
factor(STATE)NJ  6.249e-01  9.246e-02   6.759 1.48e-11 ***
factor(STATE)NM  7.445e-01  3.153e-01   2.361 0.018239 *  
factor(STATE)NV  9.514e-01  1.463e-01   6.502 8.36e-11 ***
factor(STATE)NY  6.119e-01  6.573e-02   9.310  < 2e-16 ***
factor(STATE)OH  7.073e-01  6.181e-02  11.443  < 2e-16 ***
factor(STATE)OK  1.149e+00  1.549e-01   7.414 1.34e-13 ***
factor(STATE)OR  5.062e-01  9.946e-02   5.090 3.66e-07 ***
factor(STATE)PA  6.836e-01  6.560e-02  10.420  < 2e-16 ***
factor(STATE)RI -4.530e-02  4.458e-01  -0.102 0.919081    
factor(STATE)SC  7.046e-01  5.934e-02  11.874  < 2e-16 ***
factor(STATE)SD  9.011e-01  2.229e-01   4.042 5.35e-05 ***
factor(STATE)TN  7.048e-01  8.925e-02   7.897 3.22e-15 ***
factor(STATE)TX  9.120e-01  4.909e-02  18.577  < 2e-16 ***
factor(STATE)UT  9.722e-01  1.454e-01   6.685 2.45e-11 ***
factor(STATE)VA  6.556e-01  7.249e-02   9.043  < 2e-16 ***
factor(STATE)VT  4.394e-01  3.556e-01   1.236 0.216600    
factor(STATE)WA  5.662e-01  9.788e-02   5.784 7.53e-09 ***
factor(STATE)WI  9.344e-01  1.180e-01   7.915 2.77e-15 ***
factor(STATE)WV  5.852e-01  2.860e-01   2.046 0.040787 *  
factor(STATE)WY  8.428e-01  4.170e-01   2.021 0.043317 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.179 on 8598 degrees of freedom
Multiple R-squared:  0.3109,	Adjusted R-squared:  0.3068 
F-statistic: 74.61 on 52 and 8598 DF,  p-value: < 2.2e-16
```

Appendix E: Regression Summary Table for Model 2 Fixed Effects by State only with ln(Salary)

```
Call:
lm(formula = NUM.CHILD ~ lnSALARY + factor(STATE) - 1, data = df.data)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.3962 -0.7787 -0.6582  0.3824  9.1566 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
lnSALARY         0.014843   0.002844   5.219 1.84e-07 ***
factor(STATE)AK  0.537429   0.315213   1.705 0.088236 .  
factor(STATE)AL  0.731447   0.101666   7.195 6.79e-13 ***
factor(STATE)AR  0.742869   0.096842   7.671 1.89e-14 ***
factor(STATE)AZ  0.753700   0.104062   7.243 4.77e-13 ***
factor(STATE)CA  0.713303   0.042657  16.722  < 2e-16 ***
factor(STATE)CO  0.666431   0.095999   6.942 4.15e-12 ***
factor(STATE)CT  0.561939   0.173925   3.231 0.001239 ** 
factor(STATE)DC  0.747099   0.156257   4.781 1.77e-06 ***
factor(STATE)DE  0.482704   0.372900   1.294 0.195541    
factor(STATE)FL  0.634190   0.059858  10.595  < 2e-16 ***
factor(STATE)GA  0.827643   0.065928  12.554  < 2e-16 ***
factor(STATE)HI  0.110200   0.481514   0.229 0.818982    
factor(STATE)IA  0.880168   0.093427   9.421  < 2e-16 ***
factor(STATE)ID  1.037584   0.315212   3.292 0.001000 ***
factor(STATE)IL  0.658190   0.074286   8.860  < 2e-16 ***
factor(STATE)IN  0.929376   0.071807  12.943  < 2e-16 ***
factor(STATE)KS  1.135612   0.174014   6.526 7.14e-11 ***
factor(STATE)KY  0.670556   0.101255   6.622 3.74e-11 ***
factor(STATE)LA  0.854175   0.103512   8.252  < 2e-16 ***
factor(STATE)MA  0.448661   0.100722   4.454 8.52e-06 ***
factor(STATE)MD  0.714489   0.070836  10.087  < 2e-16 ***
factor(STATE)ME  0.942514   0.251640   3.745 0.000181 ***
factor(STATE)MI  0.811251   0.056866  14.266  < 2e-16 ***
factor(STATE)MN  0.846043   0.113256   7.470 8.79e-14 ***
factor(STATE)MO  0.839387   0.075408  11.131  < 2e-16 ***
factor(STATE)MS  0.771876   0.065290  11.822  < 2e-16 ***
factor(STATE)MT  0.787088   0.372894   2.111 0.034822 *  
factor(STATE)NC  0.686399   0.054046  12.700  < 2e-16 ***
factor(STATE)ND  0.151095   0.373003   0.405 0.685431    
factor(STATE)NE  1.220321   0.125945   9.689  < 2e-16 ***
factor(STATE)NH  0.735755   0.251499   2.925 0.003448 ** 
factor(STATE)NJ  0.611159   0.092708   6.592 4.59e-11 ***
factor(STATE)NM  0.737389   0.315282   2.339 0.019367 *  
factor(STATE)NV  0.943531   0.146341   6.448 1.20e-10 ***
factor(STATE)NY  0.613049   0.065618   9.343  < 2e-16 ***
factor(STATE)OH  0.697359   0.061920  11.262  < 2e-16 ***
factor(STATE)OK  1.133839   0.155009   7.315 2.81e-13 ***
factor(STATE)OR  0.493136   0.099581   4.952 7.48e-07 ***
factor(STATE)PA  0.670128   0.065803  10.184  < 2e-16 ***
factor(STATE)RI -0.091870   0.446031  -0.206 0.836818    
factor(STATE)SC  0.693092   0.059474  11.654  < 2e-16 ***
factor(STATE)SD  0.888772   0.222972   3.986 6.77e-05 ***
factor(STATE)TN  0.694380   0.089339   7.772 8.59e-15 ***
factor(STATE)TX  0.900605   0.049352  18.249  < 2e-16 ***
factor(STATE)UT  0.962116   0.145501   6.612 4.01e-11 ***
factor(STATE)VA  0.642072   0.072725   8.829  < 2e-16 ***
factor(STATE)VT  0.425077   0.355577   1.195 0.231941    
factor(STATE)WA  0.558208   0.097961   5.698 1.25e-08 ***
factor(STATE)WI  0.922852   0.118131   7.812 6.29e-15 ***
factor(STATE)WV  0.579125   0.285995   2.025 0.042904 *  
factor(STATE)WY  0.786552   0.417243   1.885 0.059448 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.179 on 8598 degrees of freedom
Multiple R-squared:  0.3112,	Adjusted R-squared:  0.307 
F-statistic: 74.69 on 52 and 8598 DF,  p-value: < 2.2e-16
```

Appendix F: Regression Summary Table for Model 3 Fixed Effects by State with All Explanatory Variables but not Age Stratifying and Interaction

```
Call:
lm(formula = NUM.CHILD ~ AGE.REFP + SEX.REFP + AGE.SPOUSE + SEX.SPOUSE +
    lnSALARY + EDU.FATHER + EDU.MOTHER + RACE.SPOUSE + RELIGION.SPOUSE +
    RACE.REFP + RELIGION.REFP + SAMESEX + factor(STATE) - 1,
    data = df.data)

Residuals:
    Min      1Q  Median      3Q     Max
-2.4395 -0.6164 -0.2338  0.3997  8.8488

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
AGE.REFP         -0.016249   0.000875 -18.570  < 2e-16 ***
SEX.REFP          0.597840   0.032920  18.160  < 2e-16 ***
AGE.SPOUSE       -0.023839   0.001416 -16.839  < 2e-16 ***
SEX.SPOUSE        1.033507   0.045390  22.769  < 2e-16 ***
lnSALARY         -0.007986   0.002638  -3.027 0.002478 **
EDU.FATHER       -0.016972   0.010387  -1.634 0.102301
EDU.MOTHER       -0.002492   0.010615  -0.235 0.814430
RACE.SPOUSE      -0.010377   0.014256  -0.728 0.466704
RELIGION.SPOUSE   0.005979   0.002738   2.183 0.029027 *
RACE.REFP         0.044795   0.010978   4.080 4.54e-05 ***
RELIGION.REFP     0.003281   0.001751   1.874 0.061009 .
SAMESEX          -0.921942   0.154452  -5.969 2.48e-09 ***
factor(STATE)AK   0.133512   0.286238   0.466 0.640915
factor(STATE)AL   0.226367   0.113619   1.992 0.046366 *
factor(STATE)AR   0.233693   0.109773   2.129 0.033293 *
factor(STATE)AZ   0.207488   0.114663   1.810 0.070400 .
factor(STATE)CA   0.116409   0.078887   1.476 0.140078
factor(STATE)CO   0.116113   0.106501   1.090 0.275632
factor(STATE)CT   0.086866   0.169106   0.514 0.607492
factor(STATE)DC   0.136088   0.155148   0.877 0.380431
factor(STATE)DE   0.101008   0.337335   0.299 0.764620
factor(STATE)FL   0.182643   0.086501   2.111 0.034761 *
factor(STATE)GA   0.255784   0.089790   2.849 0.004400 **
factor(STATE)HI  -0.139348   0.430207  -0.324 0.746016
factor(STATE)IA   0.338349   0.106602   3.174 0.001509 **
factor(STATE)ID   0.299328   0.287229   1.042 0.297384
factor(STATE)IL   0.166214   0.094324   1.762 0.078076 .
factor(STATE)IN   0.298924   0.093225   3.206 0.001349 **
factor(STATE)KS   0.337155   0.167955   2.007 0.044736 *
factor(STATE)KY   0.196244   0.111818   1.755 0.079290 .
factor(STATE)LA   0.298742   0.114042   2.620 0.008819 **
factor(STATE)MA  -0.024722   0.112479  -0.220 0.826039
factor(STATE)MD   0.212560   0.092974   2.286 0.022265 *
factor(STATE)ME   0.640047   0.233209   2.745 0.006073 **
factor(STATE)MI   0.252352   0.083858   3.009 0.002626 **
factor(STATE)MN   0.336836   0.120718   2.790 0.005278 **
factor(STATE)MO   0.253601   0.094825   2.674 0.007500 **
factor(STATE)MS   0.229664   0.089684   2.561 0.010460 *
factor(STATE)MT   0.401911   0.337051   1.192 0.233124
factor(STATE)NC   0.155165   0.082477   1.881 0.059965 .
factor(STATE)ND  -0.561821   0.337348  -1.665 0.095869 .
factor(STATE)NE   0.484132   0.131680   3.677 0.000238 ***
factor(STATE)NH   0.203843   0.232569   0.876 0.380791
factor(STATE)NJ   0.233672   0.106890   2.186 0.028835 *
factor(STATE)NM   0.103977   0.287637   0.361 0.717744
factor(STATE)NV   0.346744   0.145731   2.379 0.017365 *
factor(STATE)NY   0.115249   0.089622   1.286 0.198498
factor(STATE)OH   0.161695   0.087020   1.858 0.063183 .
factor(STATE)OK   0.443845   0.152703   2.907 0.003663 **
factor(STATE)OR   0.057047   0.110176   0.518 0.604624
factor(STATE)PA   0.177521   0.090136   1.969 0.048931 *
factor(STATE)RI  -0.762639   0.400942  -1.902 0.057189 .
factor(STATE)SC   0.164697   0.087100   1.891 0.058673 .
factor(STATE)SD   0.431544   0.208953   2.065 0.038927 *
factor(STATE)TN   0.073921   0.104806   0.705 0.480632
factor(STATE)TX   0.320626   0.080471   3.984 6.82e-05 ***
factor(STATE)UT   0.483391   0.147699   3.273 0.001069 **
factor(STATE)VA   0.091102   0.094698   0.962 0.336060
factor(STATE)VT  -0.151991   0.321592  -0.473 0.636496
factor(STATE)WA   0.020964   0.110010   0.191 0.848869
factor(STATE)WI   0.258836   0.125236   2.067 0.038783 *
factor(STATE)WV   0.101600   0.260912   0.389 0.696987
factor(STATE)WY   0.389630   0.374781   1.040 0.298546
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.045 on 8587 degrees of freedom
Multiple R-squared:  0.4595,    Adjusted R-squared:  0.4556
F-statistic: 115.9 on 63 and 8587 DF,  p-value: < 2.2e-16
```

Appendix G: Regression Summary Table for Model 4 Fixed Effects by State with All Explanatory Variables and Age Stratifying but not Interaction

```
Call:
lm(formula = NUM.CHILD ~ AGE.REFP + SEX.REFP + AGE.SPOUSE + SEX.SPOUSE +
    lnSALARY + EDU.FATHER + EDU.MOTHER + RACE.SPOUSE + RELIGION.SPOUSE +
    RACE.REFP + RELIGION.REFP + SAMESEX + x40ROLDER.REFP + x50ROLDER.REFP +
    x60ROLDER.REFP + x40ROLDER.SPOUSE + x50ROLDER.SPOUSE +
    x60ROLDER.SPOUSE + factor(STATE) - 1, data = df.data)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1711 -0.5235 -0.1928  0.3883  8.8230

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
AGE.REFP            0.009524   0.002084   4.571 4.93e-06 ***
SEX.REFP            0.602557   0.031452  19.158  < 2e-16 ***
AGE.SPOUSE          0.027375   0.003613   7.576 3.94e-14 ***
SEX.SPOUSE          0.224875   0.065978   3.408 0.000657 ***
lnSALARY           -0.014772   0.002552  -5.789 7.32e-09 ***
EDU.FATHER         -0.019241   0.009925  -1.939 0.052586 .
EDU.MOTHER         -0.003580   0.010146  -0.353 0.724235
RACE.SPOUSE        -0.011712   0.013614  -0.860 0.389658
RELIGION.SPOUSE     0.006078   0.002616   2.324 0.020165 *
RACE.REFP           0.045765   0.010485   4.365 1.29e-05 ***
RELIGION.REFP       0.003146   0.001672   1.881 0.059962 .
SAMESEX            -1.063124   0.147915  -7.187 7.16e-13 ***
x40ROLDER.REFP     -0.086833   0.041763  -2.079 0.037629 *
x50ROLDER.REFP     -0.722731   0.047348 -15.264  < 2e-16 ***
x60ROLDER.REFP     -0.335303   0.050281  -6.669 2.74e-11 ***
x40ROLDER.SPOUSE   -0.466573   0.062730  -7.438 1.12e-13 ***
x50ROLDER.SPOUSE   -0.901259   0.069227 -13.019  < 2e-16 ***
x60ROLDER.SPOUSE   -0.431807   0.072418  -5.963 2.58e-09 ***
factor(STATE)AK    -0.572907   0.278159  -2.060 0.039463 *
factor(STATE)AL    -0.573895   0.120565  -4.760 1.97e-06 ***
factor(STATE)AR    -0.500017   0.117038  -4.272 1.96e-05 ***
factor(STATE)AZ    -0.551181   0.121410  -4.540 5.71e-06 ***
factor(STATE)CA    -0.636963   0.091939  -6.928 4.57e-12 ***
factor(STATE)CO    -0.619514   0.113740  -5.447 5.27e-08 ***
factor(STATE)CT    -0.686212   0.169609  -4.046 5.26e-05 ***
factor(STATE)DC    -0.650151   0.157815  -4.120 3.83e-05 ***
factor(STATE)DE    -0.702724   0.325822  -2.157 0.031051 *
factor(STATE)FL    -0.608096   0.098107  -6.198 5.97e-10 ***
factor(STATE)GA    -0.521352   0.100310  -5.197 2.07e-07 ***
factor(STATE)HI    -0.888081   0.414363  -2.143 0.032121 *
factor(STATE)IA    -0.448123   0.115287  -3.887 0.000102 ***
factor(STATE)ID    -0.380165   0.278904  -1.363 0.172898
factor(STATE)IL    -0.594586   0.104386  -5.696 1.27e-08 ***
factor(STATE)IN    -0.490905   0.103110  -4.761 1.96e-06 ***
factor(STATE)KS    -0.433009   0.168483  -2.570 0.010185 *
factor(STATE)KY    -0.510613   0.118934  -4.293 1.78e-05 ***
factor(STATE)LA    -0.491263   0.121403  -4.047 5.24e-05 ***
factor(STATE)MA    -0.733842   0.119658  -6.133 9.01e-10 ***
factor(STATE)MD    -0.539811   0.103136  -5.234 1.70e-07 ***
factor(STATE)ME    -0.238383   0.229685  -1.038 0.299359
factor(STATE)MI    -0.506297   0.095768  -5.287 1.28e-07 ***
factor(STATE)MN    -0.463201   0.127076  -3.645 0.000269 ***
factor(STATE)MO    -0.557609   0.104968  -5.312 1.11e-07 ***
factor(STATE)MS    -0.547186   0.100364  -5.452 5.12e-08 ***
factor(STATE)MT    -0.602344   0.327540  -1.839 0.065950 .
factor(STATE)NC    -0.621542   0.094784  -6.557 5.79e-11 ***
factor(STATE)ND    -1.323110   0.326445  -4.053 5.10e-05 ***
factor(STATE)NE    -0.371372   0.137173  -2.707 0.006796 **
factor(STATE)NH    -0.565333   0.228634  -2.473 0.013431 *
factor(STATE)NJ    -0.532870   0.114720  -4.645 3.45e-06 ***
factor(STATE)NM    -0.657447   0.280048  -2.348 0.018916 *
factor(STATE)NV    -0.432607   0.147810  -2.927 0.003434 **
factor(STATE)NY    -0.657041   0.100607  -6.531 6.92e-11 ***
factor(STATE)OH    -0.612216   0.098720  -6.202 5.85e-10 ***
factor(STATE)OK    -0.300672   0.155065  -1.939 0.052534 .
factor(STATE)OR    -0.741395   0.117821  -6.293 3.28e-10 ***
factor(STATE)PA    -0.603745   0.101221  -5.965 2.55e-09 ***
factor(STATE)RI    -1.418390   0.386422  -3.671 0.000243 ***
factor(STATE)SC    -0.589518   0.098352  -5.994 2.13e-09 ***
factor(STATE)SD    -0.439693   0.206770  -2.126 0.033492 *
factor(STATE)TN    -0.719158   0.113527  -6.335 2.50e-10 ***
factor(STATE)TX    -0.459251   0.092921  -4.942 7.86e-07 ***
factor(STATE)UT    -0.403137   0.151425  -2.662 0.007776 **
factor(STATE)VA    -0.678106   0.104576  -6.484 9.41e-11 ***
factor(STATE)VT    -1.000473   0.311619  -3.211 0.001330 **
factor(STATE)WA    -0.736332   0.117960  -6.242 4.52e-10 ***
factor(STATE)WI    -0.508000   0.130183  -3.902 9.61e-05 ***
factor(STATE)WV    -0.578027   0.254230  -2.274 0.023012 *
factor(STATE)WY    -0.343060   0.361886  -0.948 0.343167
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.9979 on 8581 degrees of freedom
Multiple R-squared:  0.5076,    Adjusted R-squared:  0.5037
F-statistic: 128.2 on 69 and 8581 DF,  p-value: < 2.2e-16
```

Appendix H: Regression Summary Table for Model 5 Fixed Effects by State with All Explanatory Variables, Age Stratifying and Interaction

```
Call:
lm(formula = NUM.CHILD ~ AGE.REFP + SEX.REFP + AGE.SPOUSE + SEX.SPOUSE +
    lnSALARY + EDU.FATHER + EDU.MOTHER + RACE.SPOUSE + RELIGION.SPOUSE +
    RACE.REFP + RELIGION.REFP + SAMESEX + x40OROLDER.REFP + x50OROLDER.REFP +
    x60OROLDER.REFP + x40OROLDER.SPOUSE + x50OROLDER.SPOUSE +
    x60OROLDER.SPOUSE + AGE.REFP * x40OROLDER.REFP + AGE.REFP *
    x50OROLDER.REFP + AGE.REFP * x60OROLDER.REFP + AGE.SPOUSE *
    x40OROLDER.SPOUSE + AGE.SPOUSE * x50OROLDER.SPOUSE + AGE.SPOUSE *
    x60OROLDER.SPOUSE + factor(STATE) - 1, data = df.data)

Residuals:
    Min      1Q  Median      3Q     Max
-2.5434 -0.5216 -0.1651  0.3482  8.7679

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
AGE.REFP                      0.042308   0.003423  12.358  < 2e-16 ***
SEX.REFP                      0.611547   0.030570  20.005  < 2e-16 ***
AGE.SPOUSE                    0.064195   0.005165  12.428  < 2e-16 ***
SEX.SPOUSE                   -0.379995   0.086683  -4.384 1.18e-05 ***
lnSALARY                     -0.020094   0.002495  -8.053 9.13e-16 ***
EDU.FATHER                   -0.020521   0.009650  -2.127 0.033482 *
EDU.MOTHER                   -0.003138   0.009864  -0.318 0.750390
RACE.SPOUSE                  -0.015192   0.013241  -1.147 0.251277
RELIGION.SPOUSE               0.006105   0.002542   2.402 0.016338 *
RACE.REFP                     0.043909   0.010193   4.308 1.67e-05 ***
RELIGION.REFP                 0.003296   0.001625   2.029 0.042520 *
SAMESEX                      -1.216151   0.144786  -8.400  < 2e-16 ***
x40OROLDER.REFP               3.235972   0.409534   7.902 3.09e-15 ***
x50OROLDER.REFP              -0.550145   0.679438  -0.810 0.418131
x60OROLDER.REFP              -1.750052   0.587298  -2.980 0.002892 **
x40OROLDER.SPOUSE             5.799920   0.584681   9.920  < 2e-16 ***
x50OROLDER.SPOUSE            -2.257980   0.960315  -2.351 0.018731 *
x60OROLDER.SPOUSE            -2.325473   0.881344  -2.639 0.008341 **
factor(STATE)AK              -1.598597   0.283411  -5.641 1.75e-08 ***
factor(STATE)AL              -1.578428   0.144201 -10.946  < 2e-16 ***
factor(STATE)AR              -1.469858   0.139718 -10.520  < 2e-16 ***
factor(STATE)AZ              -1.513487   0.143898 -10.518  < 2e-16 ***
factor(STATE)CA              -1.622521   0.121711 -13.331  < 2e-16 ***
factor(STATE)CO              -1.588173   0.138110 -11.499  < 2e-16 ***
factor(STATE)CT              -1.644178   0.184482  -8.912  < 2e-16 ***
factor(STATE)DC              -1.654328   0.174540  -9.478  < 2e-16 ***
factor(STATE)DE              -1.684140   0.327341  -5.145 2.74e-07 ***
factor(STATE)FL              -1.600031   0.126705 -12.628  < 2e-16 ***
factor(STATE)GA              -1.497376   0.128147 -11.685  < 2e-16 ***
factor(STATE)HI              -1.957192   0.411043  -4.762 1.95e-06 ***
factor(STATE)IA              -1.449499   0.139203 -10.413  < 2e-16 ***
factor(STATE)ID              -1.385597   0.282595  -4.903 9.61e-07 ***
factor(STATE)IL              -1.581820   0.131022 -12.073  < 2e-16 ***
factor(STATE)IN              -1.489454   0.130626 -11.402  < 2e-16 ***
factor(STATE)KS              -1.436498   0.183571  -7.825 5.67e-15 ***
factor(STATE)KY              -1.488913   0.141665 -10.510  < 2e-16 ***
factor(STATE)LA              -1.494253   0.144843 -10.316  < 2e-16 ***
factor(STATE)MA              -1.732860   0.142948 -12.122  < 2e-16 ***
factor(STATE)MD              -1.559267   0.130812 -11.920  < 2e-16 ***
factor(STATE)ME              -1.276770   0.238170  -5.361 8.50e-08 ***
factor(STATE)MI              -1.481144   0.125012 -11.848  < 2e-16 ***
factor(STATE)MN              -1.452559   0.148421  -9.787  < 2e-16 ***
factor(STATE)MO              -1.528712   0.131270 -11.646  < 2e-16 ***
factor(STATE)MS              -1.551707   0.129193 -12.011  < 2e-16 ***
factor(STATE)MT              -1.477526   0.328951  -4.492 7.16e-06 ***
factor(STATE)NC              -1.622655   0.124503 -13.033  < 2e-16 ***
factor(STATE)ND              -2.066909   0.328057  -6.300 3.11e-10 ***
factor(STATE)NE              -1.423407   0.157489  -9.038  < 2e-16 ***
factor(STATE)NH              -1.619664   0.238957  -6.778 1.30e-11 ***
factor(STATE)NJ              -1.512834   0.138810 -10.899  < 2e-16 ***
factor(STATE)NM              -1.556140   0.284115  -5.477 4.44e-08 ***
factor(STATE)NV              -1.456379   0.165976  -8.775  < 2e-16 ***
factor(STATE)NY              -1.645659   0.128671 -12.790  < 2e-16 ***
factor(STATE)OH              -1.562462   0.126747 -12.327  < 2e-16 ***
factor(STATE)OK              -1.243167   0.172297  -7.215 5.84e-13 ***
factor(STATE)OR              -1.707268   0.140803 -12.125  < 2e-16 ***
factor(STATE)PA              -1.581318   0.128463 -12.310  < 2e-16 ***
factor(STATE)RI              -2.276952   0.384345  -5.924 3.26e-09 ***
factor(STATE)SC              -1.591304   0.126937 -12.536  < 2e-16 ***
factor(STATE)SD              -1.435993   0.216939  -6.619 3.82e-11 ***
factor(STATE)TN              -1.689404   0.138144 -12.229  < 2e-16 ***
factor(STATE)TX              -1.451866   0.123106 -11.794  < 2e-16 ***
factor(STATE)UT              -1.351705   0.168028  -8.045 9.81e-16 ***
factor(STATE)VA              -1.679306   0.131597 -12.761  < 2e-16 ***
factor(STATE)VT              -1.973317   0.312744  -6.310 2.93e-10 ***
factor(STATE)WA              -1.724497   0.141071 -12.224  < 2e-16 ***
factor(STATE)WI              -1.437257   0.151182  -9.507  < 2e-16 ***
factor(STATE)WV              -1.473367   0.259622  -5.675 1.43e-08 ***
factor(STATE)WY              -1.336306   0.360842  -3.703 0.000214 ***
AGE.REFP:x40OROLDER.REFP     -0.086687   0.009655  -8.978  < 2e-16 ***
AGE.REFP:x50OROLDER.REFP      0.008533   0.013516   0.631 0.527848
AGE.REFP:x60OROLDER.REFP      0.029828   0.010408   2.866 0.004170 **
AGE.SPOUSE:x40OROLDER.SPOUSE -0.151313   0.013788 -10.974  < 2e-16 ***
AGE.SPOUSE:x50OROLDER.SPOUSE  0.047025   0.019120   2.459 0.013935 *
AGE.SPOUSE:x60OROLDER.SPOUSE  0.040594   0.015442   2.629 0.008584 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9696 on 8575 degrees of freedom
Multiple R-squared:  0.5355,    Adjusted R-squared:  0.5315
F-statistic: 131.8 on 75 and 8575 DF,  p-value: < 2.2e-16
```

Appendix I: Regression Summary Table for Final Model [Model 6] Fixed Effects by State with All Explanatory Variables, Age Stratifying, Interaction, and Dropping Non Statistically Significant Coefficients

```
Call:
lm(formula = NUM.CHILD ~ AGE.REFP + SEX.REFP + AGE.SPOUSE + SEX.SPOUSE +
    lnSALARY + EDU.FATHER + RELIGION.SPOUSE + RACE.REFP + RELIGION.REFP +
    SAMESEX + x40ROLDER.REFP + x60ROLDER.REFP + x40ROLDER.SPOUSE +
    x50ROLDER.SPOUSE + x60ROLDER.SPOUSE + AGE.REFP * x40ROLDER.REFP +
    AGE.REFP * x60ROLDER.REFP + AGE.SPOUSE * x40ROLDER.SPOUSE +
    AGE.SPOUSE * x50ROLDER.SPOUSE + AGE.SPOUSE * x60ROLDER.SPOUSE +
    factor(STATE) - 1, data = df.data)

Residuals:
   Min      1Q  Median      3Q     Max
-2.5498 -0.5276 -0.1636  0.3491  8.7704

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
AGE.REFP                    0.042333   0.003423  12.368  < 2e-16 ***
SEX.REFP                    0.612063   0.030556  20.031  < 2e-16 ***
AGE.SPOUSE                  0.064144   0.005159  12.434  < 2e-16 ***
SEX.SPOUSE                 -0.399323   0.084736  -4.713 2.49e-06 ***
lnSALARY                   -0.020030   0.002494  -8.030 1.10e-15 ***
EDU.FATHER                 -0.020650   0.006656  -3.103 0.001925 **
RELIGION.SPOUSE             0.006215   0.002541   2.446 0.014477 *
RACE.REFP                   0.038599   0.008921   4.327 1.53e-05 ***
RELIGION.REFP               0.003294   0.001625   2.027 0.042646 *
SAMESEX                    -1.223100   0.144507  -8.464  < 2e-16 ***
x40ROLDER.REFP              3.445771   0.202509  17.015  < 2e-16 ***
x60ROLDER.REFP             -2.509619   0.269818  -9.301  < 2e-16 ***
x40ROLDER.SPOUSE            5.851534   0.574243  10.190  < 2e-16 ***
x50ROLDER.SPOUSE           -2.489860   0.926744  -2.687 0.007231 **
x60ROLDER.SPOUSE           -2.155048   0.867130  -2.485 0.012964 *
factor(STATE)AK            -1.592611   0.283243  -5.623 1.94e-08 ***
factor(STATE)AL            -1.569744   0.144009 -10.900  < 2e-16 ***
factor(STATE)AR            -1.462476   0.139514 -10.483  < 2e-16 ***
factor(STATE)AZ            -1.502996   0.143612 -10.466  < 2e-16 ***
factor(STATE)CA            -1.615927   0.121557 -13.294  < 2e-16 ***
factor(STATE)CO            -1.580559   0.137920 -11.460  < 2e-16 ***
factor(STATE)CT            -1.635790   0.184230  -8.879  < 2e-16 ***
factor(STATE)DC            -1.644106   0.174373  -9.429  < 2e-16 ***
factor(STATE)DE            -1.674542   0.327123  -5.119 3.14e-07 ***
factor(STATE)FL            -1.590802   0.126485 -12.577  < 2e-16 ***
factor(STATE)GA            -1.488024   0.127974 -11.628  < 2e-16 ***
factor(STATE)HI            -1.965885   0.410942  -4.784 1.75e-06 ***
factor(STATE)IA            -1.439619   0.138935 -10.362  < 2e-16 ***
factor(STATE)ID            -1.379521   0.282491  -4.883 1.06e-06 ***
factor(STATE)IL            -1.575153   0.130843 -12.039  < 2e-16 ***
factor(STATE)IN            -1.482363   0.130436 -11.365  < 2e-16 ***
factor(STATE)KS            -1.426888   0.183435  -7.779 8.18e-15 ***
factor(STATE)KY            -1.481769   0.141484 -10.473  < 2e-16 ***
factor(STATE)LA            -1.485852   0.144715 -10.267  < 2e-16 ***
factor(STATE)MA            -1.724632   0.142683 -12.087  < 2e-16 ***
factor(STATE)MD            -1.553687   0.130684 -11.889  < 2e-16 ***
factor(STATE)ME            -1.266563   0.238013  -5.321 1.06e-07 ***
factor(STATE)MI            -1.472757   0.124840 -11.797  < 2e-16 ***
factor(STATE)MN            -1.445518   0.148246  -9.751  < 2e-16 ***
factor(STATE)MO            -1.522850   0.131080 -11.618  < 2e-16 ***
factor(STATE)MS            -1.543054   0.129016 -11.960  < 2e-16 ***
factor(STATE)MT            -1.467891   0.328887  -4.463 8.18e-06 ***
factor(STATE)NC            -1.615642   0.124337 -12.994  < 2e-16 ***
factor(STATE)ND            -2.056993   0.327942  -6.272 3.73e-10 ***
factor(STATE)NE            -1.417223   0.157354  -9.007  < 2e-16 ***
factor(STATE)NH            -1.610713   0.238864  -6.743 1.65e-11 ***
factor(STATE)NJ            -1.505977   0.138630 -10.863  < 2e-16 ***
factor(STATE)NM            -1.551213   0.284001  -5.462 4.84e-08 ***
factor(STATE)NV            -1.452111   0.165899  -8.753  < 2e-16 ***
factor(STATE)NY            -1.639079   0.128533 -12.752  < 2e-16 ***
factor(STATE)OH            -1.553669   0.126551 -12.277  < 2e-16 ***
factor(STATE)OK            -1.231882   0.172069  -7.159 8.78e-13 ***
factor(STATE)OR            -1.698695   0.140666 -12.076  < 2e-16 ***
factor(STATE)PA            -1.571219   0.128247 -12.251  < 2e-16 ***
factor(STATE)RI            -2.267685   0.384290  -5.901 3.75e-09 ***
factor(STATE)SC            -1.585498   0.126769 -12.507  < 2e-16 ***
factor(STATE)SD            -1.430923   0.216913  -6.597 4.45e-11 ***
factor(STATE)TN            -1.679414   0.137935 -12.175  < 2e-16 ***
factor(STATE)TX            -1.444306   0.122954 -11.747  < 2e-16 ***
factor(STATE)UT            -1.339495   0.167854  -7.980 1.65e-15 ***
factor(STATE)VA            -1.673024   0.131423 -12.730  < 2e-16 ***
factor(STATE)VT            -1.965852   0.312633  -6.288 3.37e-10 ***
factor(STATE)WA            -1.714306   0.140867 -12.170  < 2e-16 ***
factor(STATE)WI            -1.430335   0.151068  -9.468  < 2e-16 ***
factor(STATE)WV            -1.468319   0.259456  -5.659 1.57e-08 ***
factor(STATE)WY            -1.324111   0.360769  -3.670 0.000244 ***
AGE.REFP:x40ROLDER.REFP    -0.091731   0.004969 -18.459  < 2e-16 ***
AGE.REFP:x60ROLDER.REFP     0.043364   0.004544   9.544  < 2e-16 ***
AGE.SPOUSE:x40ROLDER.SPOUSE -0.152429  0.013513 -11.281  < 2e-16 ***
AGE.SPOUSE:x50ROLDER.SPOUSE  0.051482  0.018399   2.798 0.005152 **
AGE.SPOUSE:x60ROLDER.SPOUSE  0.037526  0.015179   2.472 0.013447 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9696 on 8579 degrees of freedom
Multiple R-squared:  0.5353,    Adjusted R-squared:  0.5315
F-statistic: 139.2 on 71 and 8579 DF,  p-value: < 2.2e-16
```